padarray Function Test Results

Test Case 1: Default Zero Padding

- **Input Matrix:** A = [1 2; 3 4]
- **Padding Size:** [1 1]
- **Option:** Default (zero padding)
- **Command:** padarray(A,[1 1])

Result: Matrix padded with zeros on all sides

0. 0. 0. 0.

0. 1. 2. 0.

0. 3. 4. 0.

0. 0. 0. 0.

Test Case 2: Explicit Zero Padding

- **Input Matrix:** A = [1 2; 3 4]
- **Padding Size:** [1 1]
- **Option:** "zeros"
- **Command:** padarray(A,[1 1],"zeros")

Result: Same output as zero padding

0. 0. 0. 0.

0. 1. 2. 0.

0. 3. 4. 0.

0. 0. 0. 0.

Test Case 3: Constant Padding

- **Input Matrix:** A = [1 2; 3 4]
- **Padding Size:** [1 1]
- **Option:** 5

- **Command:** padarray(A,[1 1],5)

Result: Borders filled with value 5

5. 5. 5. 5.

5. 1. 2. 5.

5. 3. 4. 5.

5. 5. 5. 5.

Test Case 4: Replicate Padding

- **Input Matrix:** A = [1 2; 3 4]
- **Padding Size:** [1 1]
- **Option:** "replicate"
- **Command:** padarray(A,[1 1],"replicate")

Result: Edge values repeated outward

1. 1. 2. 2.

1. 1. 2. 2.

3. 3. 4. 4.

3. 3. 4. 4.

Test Case 5: Circular Padding

- **Input Matrix:** A = [1 2; 3 4]
- **Padding Size:** [1 1]
- **Option:** "circular"
- **Command:** padarray(A,[1 1],"circular")

Result: Values wrapped from opposite edges

4. 3. 4. 3.

2. 1. 2. 1.

4. 3. 4. 3.

2. 1. 2. 1.

Test Case 6: Reflect Padding

- **Input Matrix:** A = [1 2 3; 4 5 6; 7 8 9]
- **Padding Size:** [1 1]
- **Option:** "reflect"
- **Command:** padarray(A,[1 1],"reflect")

Result: Matrix reflected excluding edge pixels

5. 4. 5. 6. 5.

2. 1. 2. 3. 2.

5. 4. 5. 6. 5.

8. 7. 8. 9. 8.

5. 4. 5. 6. 5.

Test Case 7: Symmetric Padding

- **Input Matrix:** Same as above 3×3 matrix
- **Padding Size:** [1 1]
- **Option:** "symmetric"
- **Command:** padarray(A,[1 1],"symmetric")

Result: Matrix reflected including edge pixels

1. 1. 2. 3. 3.

1. 1. 2. 3. 3.

4. 4. 5. 6. 6.

7. 7. 8. 9. 9.

7. 7. 8. 9. 9.

Test Case 8: Non-Square Padding

- **Input Matrix:** A = [1 2 3; 4 5 6]
- **Padding Size:** [1 3]

- **Option:** "replicate"
- **Command:** padarray(A,[1 3],"replicate")

Result: Unequal row and column padding applied

1. 1. 1. 1. 2. 3. 3. 3. 3.

1. 1. 1. 1. 2. 3. 3. 3. 3.

4. 4. 4. 4. 5. 6. 6. 6. 6.

4. 4. 4. 4. 5. 6. 6. 6. 6.

Test Case 9: Single Pixel Image

- **Input Matrix:** A = 7
- **Padding Size:** [2 2]
- **Option:** 0
- **Command:** padarray(A,[2 2],0)

Result: Center value preserved with zero border

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

0. 0. 7. 0. 0.

0. 0. 0. 0. 0.

0. 0. 0. 0. 0.

Test Case 10: No Padding

- **Input Matrix:** A = [1 2; 3 4]
- **Padding Size:** [0 0]
- **Option:** Default
- **Command:** padarray(A,[0 0])

Result: Original matrix returned unchanged

1. 2.

3. 4.

Test Case 11: Invalid Negative Padding

- **Input Matrix:** A = [1 2; 3 4]
- **Padding Size:** [-1 1]
- **Option:** "replicate"
- **Command:** padarray(A,[-1 1],"replicate")

Result: Error handled correctly - "padarray: padsize values must be non-negative."

Test Case 12: Invalid Reflect Padding

- **Input Matrix:** A = [1 2; 3 4]
- **Padding Size:** [2 2]
- **Option:** "reflect"
- **Command:** padarray(A,[2 2],"reflect")

Result: Error handled correctly - "padarray: unsupported padding type."